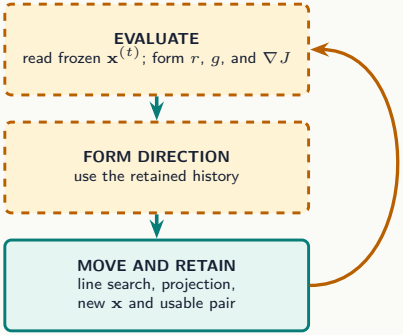
L-BFGS in one loop: evaluate, move, and remember

SYNC BEAT 1 OF 7 · STATE NOW

ITERATION BOUNDARY: retain one scalar vector $\mathbf{x}^{(t)}$ and up to ten accepted $(\mathbf{s}, \mathbf{y})$ history pairs. All check terms are recalculated.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 0 scalars
retained now: 3 / 63 configured cap

EVALUATE AT FROZEN $\mathbf{x}^{(t)}$ ⟶ **FORM A DIRECTION** ⟶ **ACCEPT A PROJECTED STEP** ⟶ **REPEAT**



Objective function in this three-bit teaching variant

$$J(\mathbf{x}) = -\frac{1}{2}\sum_{i=1}^{3} \ell_i x_i + 2 \sum_{c \in \{A,B\}} \left(1 - \prod_{j \in c} x_j\right)^2$$

One accepted update

$$\mathbf{d}^{(t)} = -\mathbf{H}_t \nabla J(\mathbf{x}^{(t)}), \qquad \mathbf{x}^{(t+1)} = \Pi_{[-1,1]^3}\left(\mathbf{x}^{(t)} + \alpha_t \mathbf{d}^{(t)}\right)$$

L-BFGS retains one current scalar per bit and accepted history pairs; its temporary check calculations are not Tanner-graph messages.

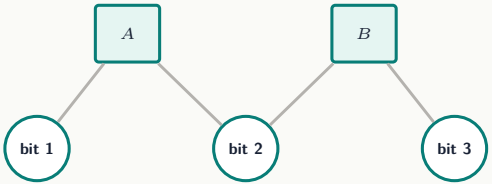
Boundary-retained count excludes fixed $\boldsymbol{\ell}$, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity

Initialize once: channel evidence and one scalar per bit

SYNC BEAT 2 OF 7 · STATE NOW
INITIAL VALUES: fixed $\boldsymbol{\ell} = -\mathbf{L}$, current $\mathbf{x}^{(0)} = \tanh(\boldsymbol{\ell}/2)$, and an empty L-BFGS history.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 0 scalars
retained now: 3 / 63 configured cap

PLACE FIXED $\boldsymbol{\ell}$ ⟶ **INITIALIZE $\mathbf{x}^{(0)}$** ⟶ **KEEP ONE SCALAR PER BIT**



$x_1 = +0.800$	$x_2 = -0.200$	$x_3 = -0.333$
$\ell_1 = +2.197$	$\ell_2 = -0.405$	$\ell_3 = -0.693$

$$\boldsymbol{\ell} = -\mathbf{L} = (+2.197, -0.405, -0.693)$$

$$\mathbf{x}^{(0)} = \tanh(\boldsymbol{\ell}/2) = (+0.800, -0.200, -0.333)$$

BOUNDARY-RETAINED NUMBERS

$\mathbf{x}^{(0)}[3]$	3 scalars
history pairs	$0/10$
configured cap	$3 + 2(3)(10) = 63$

Here $\ell_i > 0$ and $x_i > 0$ favour bit 0; the BP deck uses the opposite-sign channel LLR $L_i = -\ell_i$.

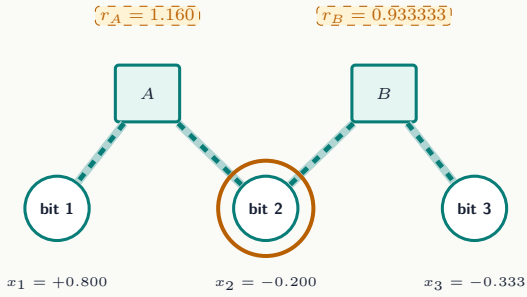
Boundary-retained count excludes fixed $\boldsymbol{\ell}$, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity

Frozen-state read: both checks evaluate their residuals

SYNC BEAT 3 OF 7 · STATE NOW
LOCKED READ: checks A and B both use the same frozen $\mathbf{x}^{(0)}$. The two residuals are temporary calculations.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 0 scalars
retained now: 3 / 63 configured cap

LOCK $\mathbf{x}^{(0)}$ ⟶ **A READS x_1, x_2** + **B READS x_2, x_3** ⟶ **FORM r_A, r_B**



$$r_c = 1 - \prod_{j \in c} x_j$$

$$r_A = 1 - (0.800)(-0.200) = 1.160,$$
$$r_B = 1 - (-0.200)(-0.333333) = 0.933333.$$

A then B is presentation order. Neither check updates $\mathbf{x}^{(0)}$.

The dashed lines show mathematical dependence on the frozen coordinates; no value travels along a Tanner-graph edge.

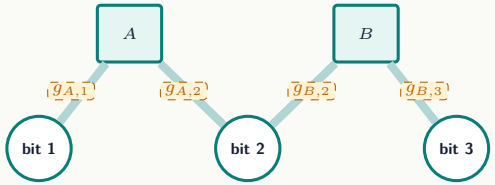
Boundary-retained count excludes fixed ℓ, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity

Check contributions: four temporary terms for the partial derivatives

SYNC BEAT 4 OF 7 · STATE NOW
ONE FROZEN SNAPSHOT: the four $g_{c,i}$ values all use $\mathbf{x}^{(0)}$. They are combined later by bit coordinate and are not retained across the boundary.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 0 scalars
retained now: 3 / 63 configured cap

READ FROZEN $\mathbf{x}^{(0)}$ AND r_c ⟶ FORM FOUR $g_{c,i}$ TERMS ⟶ KEEP $\mathbf{x}^{(0)}$ FIXED



$x_1 = +0.800$ $x_2 = -0.200$ $x_3 = -0.333$

$$g_{c,i} = -4r_c \prod_{\substack{j \in c \\ j \neq i}} x_j$$

check A	check B
$g_{A,1} = +0.928,$	$g_{B,2} = +1.244,$
$g_{A,2} = -3.712$	$g_{B,3} = +0.747$

The terms are indexed by check and coordinate only while the partial derivatives are being calculated.

These four values contribute to ∇J; they are temporary partial-derivative terms, not directed messages.

Boundary-retained count excludes fixed ℓ, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity

Aggregate by coordinate, choose a direction, then move once

SYNC BEAT 5 OF 7 · STATE NOW
START: $\mathbf{x}^{(0)}$ stays frozen while ∇J, $\mathbf{d}^{(0)}$, and the trial steps are formed. **END:** the accepted point $\mathbf{x}^{(1)}$ becomes current.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 0 scalars
retained now: 3 / 63 configured cap

ASSEMBLE ∇J $\longrightarrow$ $\mathbf{d}^{(0)} = -\nabla J$ $\longrightarrow$ **LINE SEARCH** $\longrightarrow$ **PROJECT AND MOVE**

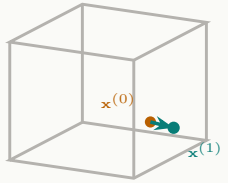
Combine all terms by bit coordinate

$$\nabla J(\mathbf{x}^{(0)}) = \begin{bmatrix} g_{A,1} - \frac{1}{2}\ell_1 \\ g_{A,2} + g_{B,2} - \frac{1}{2}\ell_2 \\ g_{B,3} - \frac{1}{2}\ell_3 \end{bmatrix} = \begin{bmatrix} -0.170612 \\ -2.264823 \\ +1.093240 \end{bmatrix}$$

$\mathbf{H}_0 = \mathbf{I}$ because the history is empty, so

$$\mathbf{d}^{(0)} = -\nabla J = (+0.170612, +2.264823, -1.093240).$$

$\alpha = 1$: $J = 6.791560$ reject $\alpha = 0.5$: $J = 5.600661$ reject
$\alpha = 0.25$: $J = 2.881689 < 3.398461$ **accept**



$[-1, 1]^3$

$$\mathbf{x}^{(1)} = \Pi\left(\mathbf{x}^{(0)} + 0.25\mathbf{d}^{(0)}\right)$$
$$= (+0.842653, +0.366206, -0.606643)$$

The accepted $\alpha = 0.25$ trial is inside the box, so the projection changes nothing.

Boundary-retained count excludes fixed ℓ, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity

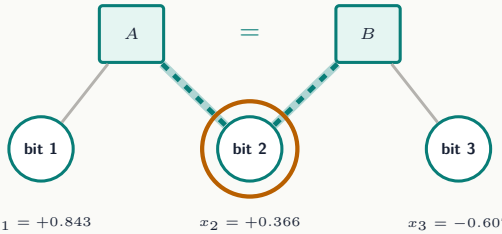
Next evaluation: both checks still read one bit-2 scalar

SYNC BEAT 6 OF 7 · STATE NOW
FORM THE FIRST HISTORY PAIR: start with $\mathbf{x}^{(1)}$. Both checks read the same $x_2^{(1)}$ to assemble $\nabla J(\mathbf{x}^{(1)})$; only then can the usable pair be retained.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 0 → 6 scalars
retained now: 3 → 9 by beat end / 63 configured cap

KEEP $\mathbf{x}^{(1)}$ (3) ⟶ BOTH CHECKS READ $x_2^{(1)}$ AND FORM $\nabla J(\mathbf{x}^{(1)})$ ⟶ FORM AND RETAIN THE PAIR (9) ⟶ FORM $\mathbf{d}^{(1)}$

A reads $x_2^{(1)} = +0.366$ B reads $x_2^{(1)} = +0.366$



1. Finish the new evaluation
$\nabla J(\mathbf{x}^{(1)}) = (-2.111, +0.838, -1.444)$

2. Only now form the history pair
$\mathbf{s}^{(0)} = (+0.043, +0.566, -0.273)$
$\mathbf{y}^{(0)} = (-1.941, +3.103, -2.537)$
$\mathbf{y}^{(0)} = \nabla J(\mathbf{x}^{(1)}) - \nabla J(\mathbf{x}^{(0)})$

3. Test, retain, then use the pair
$(\mathbf{s}^{(0)})^T \mathbf{y}^{(0)} = 2.367356 > 10^{-12}$
$\mathbf{d}^{(1)} = (+0.170, -0.055, +0.110)$

Unlike BP's two recipient-specific Q values from bit 2, L-BFGS retains one x_2 value and both checks read that value.

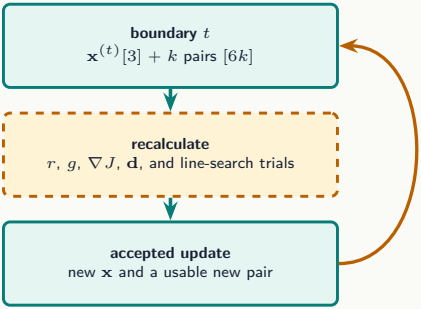
Boundary-retained count excludes fixed ℓ, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity

What crosses an L-BFGS iteration boundary

SYNC BEAT 7 OF 7 · STATE NOW
AFTER THE FIRST PAIR: $\mathbf{x}^{(1)}$, $\mathbf{s}^{(0)}$, and $\mathbf{y}^{(0)}$ cross the boundary. Residuals, check terms, the gradient, and the direction are recalculated.

RETAINED NUMBERS
current $\mathbf{x}$: 3 history: 6 scalars
retained now: 9 / 63 configured cap

KEEP $\mathbf{x}$ AND ACCEPTED HISTORY ⟶ **RECALCULATE TEMPORARY TERMS** ⟶ **ACCEPT THE NEXT UPDATE**



RETAINED AFTER THE FIRST ACCEPTED PAIR

$\mathbf{x}^{(1)}[3]$	3 coordinate values
$\mathbf{s}^{(0)}[3]$	3 history values
$\mathbf{y}^{(0)}[3]$	3 history values
retained now	**9 scalar values**

GENERAL BOUNDARY COUNT
$3 + 6k$ retained scalars for k history pairs
$0 \le k \le 10$; configured cap $= 3 + 6(10) = 63$
fixed ℓ, graph data, and temporary arithmetic excluded

Check A then check B is only presentation order: both evaluations use the same frozen iterate.

The retained state is indexed by bit coordinate and update history, not by Tanner-graph edge. In this constructed teaching variant, the displayed J is three times the anchored channel-rung objective and has no binary term; it is a mechanism tutorial, not performance evidence.

Boundary-retained count excludes fixed ℓ, graph data, and temporary arithmetic; synchronized beats do not imply equal arithmetic, storage, or complexity